

Week 4 — Confounding: what to adjust for, how to decide, and what the number can tell you

Duration: ~2.5 h (Tue 29 Sep, before the Thu 1 Oct lab). Deck: `lectures/slides/Week4_confounding_slides.qmd` — **obesity (BMI) as the only exposure**, the paper's four BMI $\times$ waist groups kept for Week 5. Its numbers come from `lectures/analysis/Week4_bmi_exposure.R`.

The deck and this plan. This plan was written for an earlier deck that ran on the paper's four phenotype groups; the deck now uses obesity vs not obese throughout, with the same figure style, Parts and plan-slide numbers. It covers **Plan Slides 1–18 and 22**; the Lab 3 bridge (**Plan Slides 19–21**) is not in the deck. Every deck slide's speaker notes name its plan slide.

Two analyses, labelled. The deck's numbers belong to the **lecture's exposure analysis**: obesity vs not obese, with every hazard ratio on the same 5,911 rows, the pre-exposure set L (age, sex, race/ethnicity, current smoking, sedentary time) as its main adjustment and the paper's Model 1 and Model 2 shown beside it. This plan's phenotype-group numbers (IV vs III, IV vs I) belong to the **paper replication**: the paper's Model 1 covariates on 5,939 rows, from the P2 and P4 exemplars; its Table 1 and full-file numbers are on 6,371 rows. The same IV-vs-III contrast differs between the two mainly because the adjustment set differs, not the rows: crude it is 2.34 on 5,939 rows and 2.31 on 5,911, but Model 1 gives 1.05 and set L 1.09 (P2 exemplar; `lectures/analysis/Week6_design_aware.json`). Every hazard ratio in this plan and the deck is unweighted, except the Week 6 pointer on Slide 22.

Timing. Per-slide markers below sum to **147 min including the break**, against a 150-minute cap. Without Plan Slides 19–21 (15 min) the deck runs about **132 min**. **If you retime a slide, retime it in the deck's notes too.** See *If you are running behind* at the foot of this file.

Written for SPPH students, and matched to EpiMethods. The deck uses plain language (every term defined on first use), EpiMethods' own notation (A, Y, L, U; $Y(A=1)$, $Y(A=0)$), and the section names and order of the EpiMethods *Concepts* page for the Causal roles chapter. Each concept slide ends with a “**Read more**” line naming the matching *Concepts* section, EpiMethods tutorial or video, and the key paper; the deck closes with a *Where to read more* slide and the full reference list.

The lecture is one argument. **On the same 5,939 rows, our locked hazard ratio (IV vs III, unweighted) moves from 2.34 to 1.05 when the paper's Model 1 covariates enter. What moved it?** There are three kinds of answer, and the lecture takes them in order: *which* variables were adjusted and what role each plays (the DAG, before the break); *how* those variables got chosen (the empirical criteria, and the data-driven methods that should not be used for this); and *what the effect measure itself does* on adjustment (collapsibility, after the break). A final block puts **Thursday's actual L3 exercise** on screen, so the lab reads as the mechanics of the second half rather than as a change of subject.

Every DAG is drawn. Each structural claim in this lecture has a figure in `lectures/img/w4_*.png`, drawn in one visual language throughout: - **Colour = role**: exposure blue, outcome vermillion, confounder orange, mediator green, collider purple, any other variable grey. - **Outlines and shapes**: a **dashed outline** means *not measured*. A **square** node is conditioned on — adjusted, stratified, restricted or selected. - **Arrows and bands**: a **grey arrow marked “?”** is the effect we want. A highlighted band marks a causal path (solid green) or an open non-causal path (dashed red).

The DAGs behind Slides 8, 17 and 19 are printed as **dagitty** code in the appendix, so students can paste them into dagitty.net and check every adjustment-set statement themselves.

Row-set discipline. Every quantitative slide names the rows it quotes. **Four** sets carry today's hazard ratios and counts, nested, and they are not interchangeable: - the **full reconstructed analytic file**, **N = 6,371** — Table 1, group sizes, death counts; - the **locked domain**, **N = 6,048** — the subset with a valid fasting weight; - its **matched complete-case subset**, **N**

= **5,939** — the 6,048 less 109 people missing cancer status; this plan’s phenotype-group crude-versus-Model-1 hazard ratios are fitted here, so crude and adjusted differ for one reason rather than two; - the deck’s **5,911** — the 6,048 less 137 people missing a covariate of the deck’s largest models (Model 2 plus race/ethnicity); every hazard ratio on the deck, and Slide 17’s, is fitted on these same rows, for the same reason.

6,371 — 5,939 is not the complete-case loss; 109 is (137 for the deck’s rows). Slide 7’s collider panels use two earlier funnel sets: the 14,170 fasting-subsample adults with FLI computable, and the 6,052 of them with $FLI \geq 60$.

What this week does NOT do. Whether an effect *differs* across subgroups, including whether WHtR’s effect differs between the obese and the non-obese, is effect modification (Week 5 and P3). Survey weights are Week 6; the missing-data mechanics (and L3’s complete-case step) are Week 7. Formal mediation analysis is not in this course. Today the phenotype is one exposure with one named contrast.

Learning objectives

1. State the **causal question** an adjusted estimate is meant to answer, as opposed to a prognostic one, and the three conditions under which adjustment answers it: **conditional exchangeability, positivity, consistency**.
2. Read a DAG: tell **causal** paths from **back-door** paths and from paths blocked at a collider; recognise the **fork, chain and collider**; and use the back-door rule to find a valid adjustment set — or to show that none exists with the variables measured.
3. Assign each covariate a **role** — confounder, proxy, mediator, collider, pre-exposure collider (M-bias), instrument, outcome-only cause — and state the **action** each role implies, including when **selection into the cohort** is itself conditioning.
4. When the DAG is incomplete, apply the **empirical criteria** (pre-treatment, common cause, disjunctive cause, modified disjunctive cause); and say why **change-in-estimate, stepwise significance and prediction-tuned machine learning** are the wrong tools for choosing confounders.
5. Explain **collapsibility**: why an OR or HR moves on adjustment with no confounding present while an RD or RR does not; distinguish **conditional** from **marginal** effects; and describe how **standardisation (g-computation)** produces a marginal one.
6. Use Table 1 to **describe and diagnose** (sparse cells, positivity, label mismatches) rather than to select, and report adjusted results without committing the **Table 2 fallacy**.
7. Recognise the four MASLD phenotype groups as a **joint exposure**, and say what that implies for the adjustment set.

Continuity

- **Last week (Wk 3 → L2 / P1):** the cohort was built and locked — eligibility, the $FLI \geq 60$ case definition, a typed funnel. M0 memos are in; the paper is locked.
- **This week:** the contrast *inside* that cohort — what distorts it, what adjustment does to it, and what the number that comes out can and cannot mean.
- **Sets up: L3** on Thursday (crude, conditional and marginal OR / RR / RD on the RHC data), **P2** (a Table 1, a variable-role table and a crude-versus-adjusted comparison on your own paper) and **Week 5** (effect modification).

The division of labour, stated on Slide 1 and again on Slide 19

	Teaches	On which data
Tuesday (this lecture)	the judgement — which variables, why, and what the effect measure does	MASLD (Kueh 2026)

	Teaches	On which data
Thursday (L3)	the mechanics — Table 1, three GLMs, crude / conditional / marginal, standardisation	RHC (right-heart catheterisation, EpiMethods confoundingE)
P2	both, transferred	the student’s locked paper

Say this out loud, twice. L3 is not MASLD. It runs on the RHC data, a binary outcome, which is precisely what makes collapsibility visible: the lab estimates one contrast as an OR, an RR and an RD, three ways each. An earlier version of this lecture told students that L3 would build a MASLD Table 1 and fit crude and age-adjusted Cox models; L3 does neither, and a student waiting for MASLD to appear on Thursday loses the first half of the lab.

Frame · 10 minutes across the slides below

Slide 1 — Today’s question, where we are, and the words we will use (6 min)

- Open with the question in plain words: non-obese adults with more waist fat died about twice as often; after adjustment for the paper’s Model 1 covariates the difference almost disappears — why?
- The arc: **Replicate** → **Interrogate** → **Improve** → **Defend**; Interrogate starts today. Show the learning objectives in one breath.
- Walk the Tue / Thu / P2 table. First of the two places L3’s different dataset is named.
- **The running example on one slide:** who (NHANES adults with MASLD, $FLI \geq 60$), the 2×2 of BMI and WHtR that makes Groups I–IV, the outcome (death, HR), and our comparison (IV vs III).
- **Notation shared with EpiMethods:** A, Y, L, U and $Y(A=1)$, $Y(A=0)$; what “adjust for / condition on” and “crude” mean. Students who meet these words once, slowly, follow the rest of the lecture.

Slide 2 — The number that moved (4 min)

- **Locked contrast, IV vs III, matched 5,939 rows, unweighted:** crude HR **2.34 (1.72–3.18)** → Model 1 **1.05 (0.77–1.45)**.
- For orientation, the replication contrast on the same rows: IV vs I, **16.05** → **3.81**. On the full 6,371-row file the reproduction gives **15.14** → **3.29** (crude on every row, Model 1 on the rows complete on its covariates), against the paper’s **15.13** → **3.155**; the gap from 16.05 → 3.81 is the row set, not the reproduction.
- Ask the room: *what could make a hazard ratio move like that?* Take three or four answers, write them on the board, and **do not evaluate them**. Slide 13 lists five mechanisms, including one nobody suggests; Slide 15 demonstrates it.
- The complete-case restriction is already ruled out, and say so: crude on the 6,048-row locked domain is 2.33; on its 5,939 complete cases, 2.34.

Part 1 — What question, and what would answer it · 12 minutes across the slides below

EpiMethods Concepts: “The Epistemological Divide”; “The Counterfactual Framework”.

Slide 3 — Explain or predict? (3 min)

- The same data can serve two questions. **Prognostic:** *which non-obese adults with MASLD are at higher risk of death?* **Causal:** *would a non-obese adult with MASLD live longer if their central adiposity were lower?*
- Prediction rewards any variable that tracks death, including downstream disease. Causal estimation punishes adjusting for downstream disease. The syntax + **covariate** is identical; the rules are opposite.

- The paper’s abstract says the phenotype “*was independently associated with higher all-cause mortality*”, and its conclusion that it “*confers the poorest survival*” — causal-sounding language on a model built like a prognostic one. Do not adjudicate it here; name the gap.
- Consequence for the second half: tools optimised for prediction (stepwise, AIC, LASSO) optimise the wrong target for today’s question.

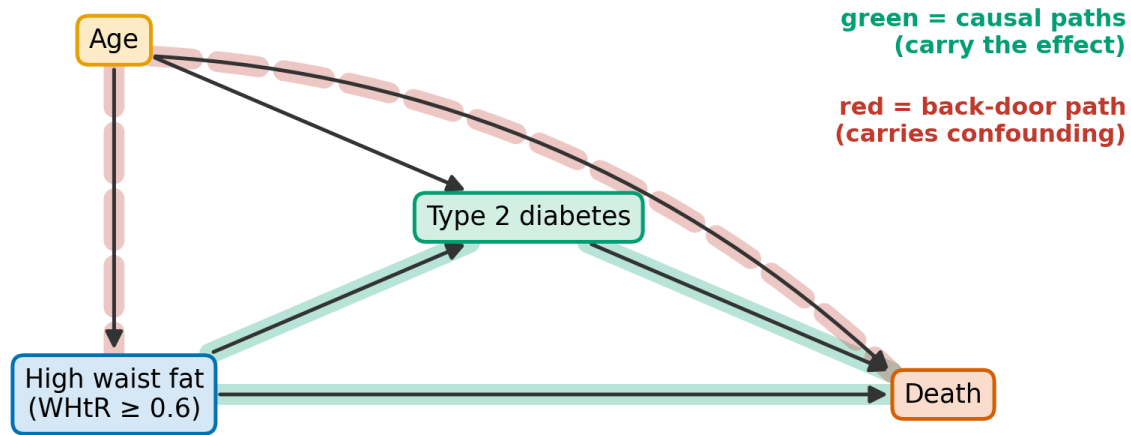
Slide 4 — Potential outcomes, three conditions, and why the exposure is tricky (9 min)

- Notation, as in EpiMethods: **A** exposure, **Y** outcome, **L** measured covariates, **U** unmeasured. Each person has **Y(IV)** and **Y(III)**: the outcome they would have in each group.
 - Put up three hypothetical rows — person, observed group, Y(IV), Y(III) — with a ? in every unobserved cell. The individual effect is never observed: causal inference is a missing-data problem. The target becomes an average, the **ATE**, $E[Y(IV) - Y(III)]$.
 - Association equals causation only if the groups are **exchangeable**. A trial buys that by randomising. We can only hope for **conditional exchangeability** given L — and it is untestable.
 - **Positivity**: within every level of L, each group must actually occur — it is about $P(\text{group} \mid \text{kind of person})$, not $P(\text{kind of person} \mid \text{group})$. On the full file ($N = 6,371$): among **obese women**, 2,550 have high waist fat and only **60 (2%)** lower waist fat, so comparisons against the paper’s Group I say almost nothing about women. Among **non-obese women**, 114 of 387 (29%) are in Group III — fine for our comparison.
 - **Say the distinction out loud**: women are 14.5% of Group III but 44.9% of Group IV. That is **imbalance**, which makes sex a **confounder** to adjust for — not a positivity problem. An earlier draft of this plan mixed the two up.
 - **Consistency**: “being in Group IV” must mean one thing. It does not — a waist can change by diet, exercise, drugs or illness, with different futures (Hernán & Taubman 2008). This is one reason the obesity-paradox literature does not settle.
 - **Why body shape is a tricky exposure (one slide)**. *What If* (Hernán & Robins, ch. 3, “Consistency: first, define the counterfactual outcome” and “The target trial”) uses obesity as its own worked example:
 - “had they not been obese” has many versions (diet, exercise, surgery, drugs, illness), each with its own effect;
 - obesity is a history, not a moment;
 - comparing obese with non-obese people implicitly emulates an impossible trial that reshapes people instantly.
- The fix is to ask about an intervention (their example: lose 5% of BMI a year from age 40) and emulate that target trial. Say why we keep the paper’s exposure anyway: it makes every bias visible.

Part 2 — Drawing the assumptions · 37 minutes across the slides below

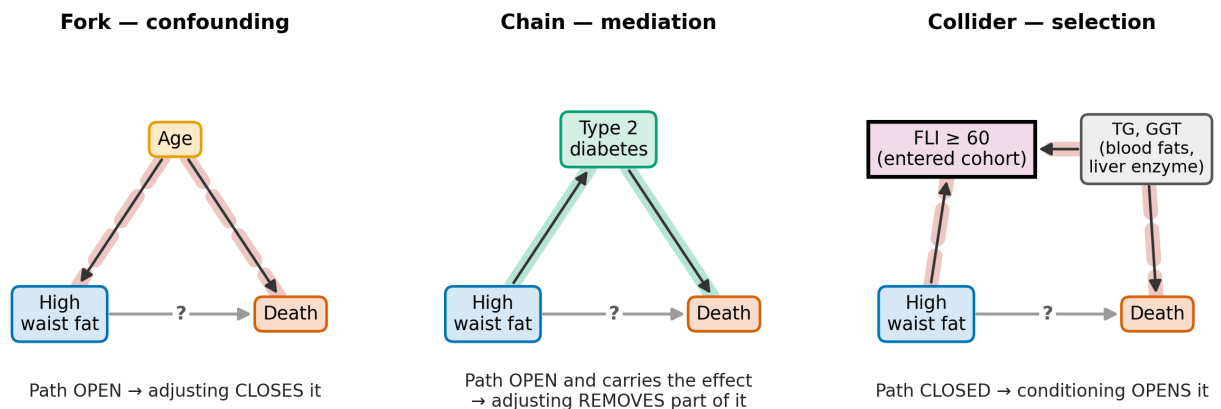
EpiMethods Concepts: “*Structural and Knowledge-Based Selection Techniques*”; “*Simpson’s Paradox*”.

Slide 5 — Reading a DAG (6 min)



- Nodes are variables; an arrow is a direct causal effect; *acyclic* is time running one way. **The absent arrows are the strongest assumptions:** here, that nothing else causes both WHtR and death.
- A **causal path** follows the arrows *forward* the whole way from exposure to outcome. A **back-door path** starts with an arrow *into* the exposure. A path that leaves the exposure but meets a collider ($\rightarrow X \leftarrow$) is non-causal, and closed unless you condition on X — Slide 6.
- The goal: block every back-door path, leave causal paths open, and open no new path by what you condition on. dagitty.net lists the paths and the minimal adjustment sets.

Slide 6 — Three building blocks, drawn in MASLD (9 min)



Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

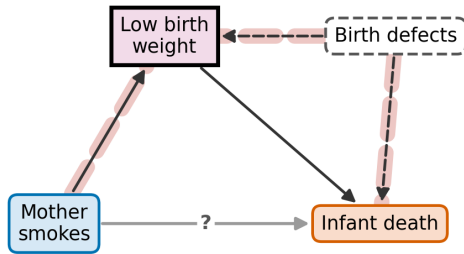
- Fork (Age):** open by default; conditioning closes it $\rightarrow$ adjust.
- Chain (type 2 diabetes):** open by default and it *carries* the effect; conditioning removes part of what you are trying to estimate $\rightarrow$ do not adjust, for a total effect.
- Collider (FLI ≥ 60):** blocked by default; conditioning *opens* it. FLI is computed from BMI, waist, triglycerides and GGT.
- Say it slowly: “conditioning” includes adjustment, stratification, restriction and selection.** This cohort was selected on FLI ≥ 60 before anyone fitted a model.
- Could adjustment undo it? Adjusting for TG and GGT would close the path the selection opened —

but TG and GGT are plausibly **effects** of central fat, so that trades selection bias for mediator and collider bias. Week 5 (Slide 18) and M1 return to this entanglement.

- One-minute check: *which structure is “hypertension”?* Take answers, then hold them — the answer depends on timing, and Slide 8 shows why the data cannot give it.

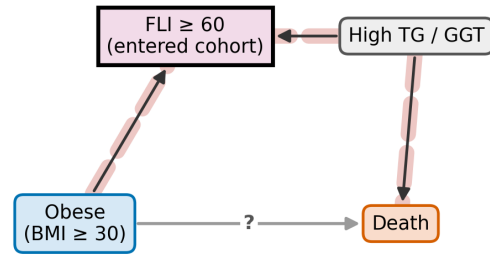
Slide 7 — A collider you can see, and paradoxes that are colliders (7 min)

The birth-weight paradox



Studying only low-birth-weight babies = conditioning on it

The obesity paradox in this MASLD cohort

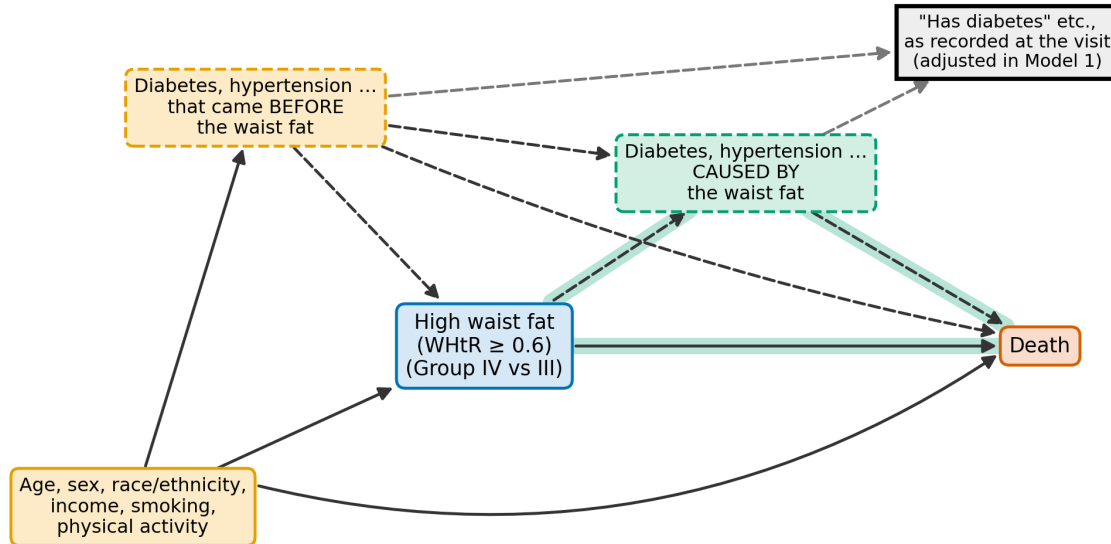


Studying only people with FLI ≥ 60 = conditioning on it

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

- **Start with a collider you can see (real data, `w4_collider_scatter.png`).**
 - Among the 14,170 fasting-subsample adults with FLI computable, higher BMI goes with higher triglycerides (Spearman $\rho = +0.28$; median TG 116 obese vs 89 not).
 - Among the 6,052 with FLI ≥ 60 the link reverses ($\rho = -0.30$; 126 vs 174). A low-BMI person stays in the cohort only if their TG is high.
 - Point at the empty lower-left corner of the right panel: that is the selection boundary.
 - Numbers from `lectures/analysis/Week4_bmi_exposure.R`.
- **Simpson’s paradox:** an association in the whole population reverses in every subgroup. The data cannot tell you which to believe; the causal structure does. Stratify on a confounder; **do not** stratify on a collider.
- **The birth-weight paradox** (Hernández-Díaz, Schisterman & Hernán 2006): among low-birth-weight infants, maternal smoking looks protective. Low birth weight is a collider of smoking and unmeasured birth defects.
- **The obesity paradox inside a disease cohort** has the same shape, and is often called *index-event bias*. A non-obese adult reaches FLI ≥ 60 mostly through high TG / GGT. In the paper’s own Model 1, Group III (non-obese) has 3.175 times the hazard of Group I (obese). FLI selection is one candidate explanation; confounding by age and by prior illness are others. **Naming a candidate is not demonstrating one.**
- **The same collider touches the locked contrast.** Within BMI < 30 , a low-WHtR adult also needs more TG / GGT to reach the threshold. On the locked rows, Group III’s median triglycerides and GGT are both higher than Group IV’s. If those carry mortality risk, IV vs III is pushed **downward** — a named bias of known direction on our own estimand.
- Keep the two halves of the headline apart. **Descriptively**, Group IV has the poorest observed survival — 19.5% died against 9.6% in Group III (paper) — and the paper reports that accurately. **Comparatively**, on the locked contrast (5,939 rows, unweighted) the Model 1 HR is **1.05 (0.77–1.45)**, and the paper’s own Model 1 implies about **0.99** for IV vs III (3.155 / 3.175, on the paper’s rows).

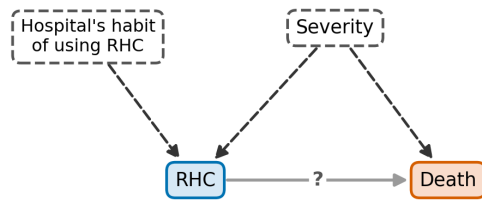
Slide 8 — A working DAG for the locked contrast (10 min) · do not drop



- Walk it left to right. Pre-exposure causes (age, sex, race/ethnicity, income, smoking, physical activity) point into both waist and death. Cardiometabolic disease appears **twice**, both unmeasured as such: disease that came *before* the central fat (a confounder) and disease *caused by* it (a mediator).
 - **Ignoring selection, the valid set** is the pre-exposure causes plus disease-before.
 - **The measurement problem is the lesson.** Model 1 enters each disease as **one composite indicator** — a self-reported past diagnosis *or* a measurement taken at the same visit that measures the waist (P2's Trap 1). Nothing records whether the disease came before or after the central fat, so the indicator is a **blend of both disease nodes**. Adjust for it, and three things happen at once:
 - some confounding is closed;
 - some of the effect is removed;
 - a new path opens: WHtR → disease-after ← disease-before → death.
- Leave it out, and disease-before stays open. Inside the FLI ≥ 60 cohort, Slide 6's TG / GGT path would also need blocking. **No valid set exists with the measured variables** — the appendix has the DAG to check it.
- Now place the paper's models. **Model 1** conditions on age, sex and the blend (plus cancer and medication use); it omits race/ethnicity and income. **Model 2** adds smoking and sedentary time. Under this DAG, neither is the valid set, and neither is clearly the worse one.
 - This is why the P2 worked example says it **cannot say which estimand Model 1 targets**. The honest move is to state the DAG, name what each model conditions on, and report both with the reading each requires. Choosing the one whose number you prefer is not an option.
 - Room exercise (2 min): *name one arrow on this figure you would dispute, or one you would add.* Record two; P2's DAG sketch is where they go.

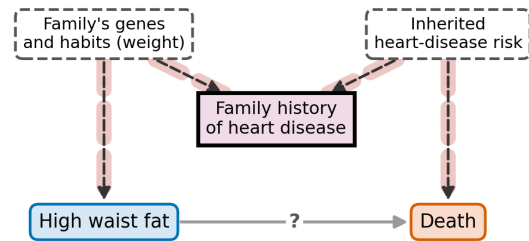
Slide 9 — More roles, and the action each implies (5 min)

Instrument (from Thursday's lab)



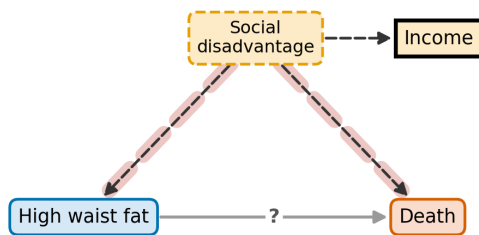
Causes the exposure only → don't adjust

M-bias — a collider measured before exposure



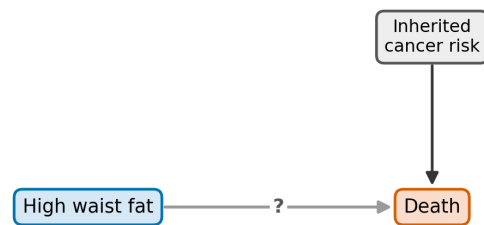
Adjusting it opens the red path (arrows drawn for illustration)

Proxy of an unmeasured confounder



Adjusting income PARTLY closes the red path

Outcome-only cause



Not a confounder: adjusting adds no bias

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

Role	Structure	Adjust?	Why
Confounder	$A \leftarrow L \rightarrow Y$	Yes	closes a back-door path
Proxy of an unmeasured confounder	$A \leftarrow U \rightarrow Y, U \rightarrow P$	Usually yes	partly closes it; residual confounding remains
Mediator	$A \rightarrow M \rightarrow Y$	No , for a total effect	removes effect; and if $M \leftarrow U \rightarrow Y$, opens a collider path
Collider, or its descendant	$A \rightarrow C \leftarrow U \rightarrow Y$	No	opens a non-causal path
Pre-exposure collider (M-bias)	$A \leftarrow U1 \rightarrow C \leftarrow U2 \rightarrow Y$	No, if it is only that; if it might also confound, adjusting is usually the safer mistake	M-bias tends to be small
Instrument	$Z \rightarrow A$ only	No	removes no bias; widens the CI; can amplify bias from U (Z-bias)
Outcome-only cause	$R \rightarrow Y$ only	Optional	RD / RR: precision. OR / HR: the number changes (Slide 15)

- Three of the four panels are MASLD:

- **Family history of CHD** is measured, pre-exposure and in the paper’s Table 1 — a plausible M-structure, and exactly what the pre-treatment rule would adjust.
- **Income** is a proxy for structural disadvantage.
- The instrument panel uses L3’s world. A hospital’s habit of using RHC *would* be an instrument **if** hospital practice affected death only through RHC — an assumption worth disputing. If it fails, the “instrument” is a confounder.

Part 3 — When the DAG runs out · 8 minutes across the slide below

EpiMethods Concepts: “Empirical Criteria for DAG-Deficient Scenarios”.

Slide 10 — Empirical criteria (8 min)

- **Pre-treatment:** adjust for everything measured before the exposure. It avoids mediators, but walks into M-bias and adjusts instruments. L3 applies this idea to a **pre-chosen list** — + . adjusts for every column the exercise kept, not everything recorded before RHC.
- **Common cause:** only variables known to cause both. Too strict: anything uncertain gets dropped, and omitting a real confounder usually costs more than including a harmless extra (EpiMethods’ reason). It can even adjust for nothing when a measured set would have worked: if a cause of the exposure (C1) and a cause of the outcome (C2) share an unmeasured parent, the back door $A \leftarrow C1 \leftarrow U \rightarrow C2 \rightarrow Y$ is open; adjusting for **either** C1 or C2 would block it, but neither is a common cause (VanderWeele & Shpitser 2011).
- **Disjunctive cause** (VanderWeele & Shpitser 2011): pre-exposure causes of the exposure **or** the outcome **or** both. Robust to not knowing which.
- **Modified disjunctive cause** (VanderWeele 2019): the same, then **drop known instruments** and **add proxies** for unmeasured common causes.
- The table’s first four rows are on the slide already filled in; **take the room’s answers on the last three**, which are the point:

Candidate	Plausibly before the exposure?	Cause of WHtR?	Cause of death?	Modified disjunctive
Age, sex	yes	yes	yes	in
Race/ethnicity, income	yes	plausibly (as proxies for structural factors)	yes	in
Smoking, physical activity	largely	yes	yes	in
Alcohol	largely	plausibly	yes	in — and an input to FLI via GGT
T2DM, hypertension, dyslipidaemia, CKD, MI	unknown	both directions	yes	cannot be placed
Cancer	unknown	both directions (weight loss)	yes	cannot be placed
Medication use	follows diagnosis	inherits the disease’s problem	—	cannot be placed

- Every criterion needs **time order**, and a single NHANES visit does not supply it for prevalent disease. The criteria reproduce Slide 8’s problem honestly rather than solving it.

Slide 11 — BREAK (10 min)

- Leave Slide 8's DAG on screen. Restart prompt: *the number moved from 2.34 to 1.05. Which of the board's guesses could you now rule out, and which could you not?*

Part 4 — How covariates get chosen in practice, and what the effect measure does · 38 minutes across the slides below

EpiMethods Concepts: “Modelling criteria for variable selection”; “Collapsibility and the Choice of Effect Measure”.

Slide 12 — Table 1 describes; it does not select (5 min)

- Four columns, full file (N = 6,371), from the reproduced Table 1 (deaths from the reproduced Table 2):

	I	II	III	IV
n	391	4,585	787	608
Deaths, n (%)	5 (1.3)	389 (8.5)	74 (9.4)	118 (19.4)
Age, mean	40.1	50.9	49.1	61.5
Female, %	15.3	55.6	14.5	44.9
Type 2 diabetes, %	10.7	27.4	13.9	30.8

- Read IV against III as **magnitudes**. Age: $(61.5 - 49.1) / 14.6 \approx$ a **0.85 SD** gap. Female: 44.9% against 14.5%. These rows say where to look; the DAG decides what they are.
- What Table 1 is *for*:
 - **Sparse cells**. Group I has 5 deaths, and 1 cardiovascular death, so the unadjusted cardiovascular IV-vs-I HR runs from 2.27 to 123.47 (reproduced, full file).
 - **Positivity**. Only 60 obese women have a lower WHtR, against 2,550 with a high one.
 - **Label mismatches**. The “< 25” BMI header from Week 3, where the Methods say 30; the income percentages P2 finds that disagree with their own counts.
- **p-values in Table 1** shrink as N grows, whatever the bias. Reproduce the paper's p column (P2 asks for it), and read standardised differences instead.
- **For Thursday**: L3's Table 1 is stratified by **death**, because that is the exercise's target table. In P2, stratify by **exposure**. Ask aloud what question each version answers.

Slide 13 — Change-in-estimate (7 min)

- The procedure: add a covariate; if the exposure coefficient moves by more than 10%, call it a confounder and keep it. Purposeful selection's “confounding check” is the same idea.
- On MASLD it fires wholesale: IV vs III, crude 2.34 → Model 1 1.05 on the same 5,939 rows, is a 55% move. Now list what can produce a move, and return to the board from Slide 2:
 1. **Confounding removed** — age, sex.
 2. **Effect removed** — the mediator half of the blend.
 3. **Bias added** — adjusting the blend opens WHtR → disease-after ← disease-before → death (Slide 8).
 4. **Non-collapsibility** of the hazard ratio — the one nobody suggests (Slide 15). It moves estimates *away* from 1, so it cannot be why this one fell toward 1.
 5. **Different rows** — ruled out here: 2.33 on the locked domain against 2.34 on its complete cases.
- Change-in-estimate cannot tell these apart; it is blind to causal structure. It is safer with an RD or RR, which have no fourth term, but no better at the other three.

Slide 14 — Significance-based and algorithmic selection (4 min)

- **Stepwise by p-value.**
 - Confounding is not a significance test: a strong confounder can be “non-significant” in a modest sample.
 - The final confidence intervals ignore the search that produced the model, so they are too narrow (post-selection inference).
 - It optimises fit. Run on the MASLD file, it would be expected to keep the cardiometabolic blend, because those variables predict death strongly — exactly the ones Slide 8 could not place.
- **Purposeful selection** (Hosmer & Lemeshow; Bursac et al. 2008): a $p < 0.25$ screen, then a change-in-estimate check. Better than raw stepwise, but still blind to causal structure — its check retains a collider, because conditioning on one moves the estimate.
- **Machine learning.** LASSO and random forests are prediction tools. Their legitimate place in causal work is the prediction sub-tasks: the propensity-score model and the outcome model.
 - **Post-double-selection** (Belloni et al. 2014) implements the disjunctive union: a LASSO for the outcome, another for the exposure, and adjust for both lists.
 - Outcome-adaptive LASSO, hdPS and C-TMLE are other confounder-oriented selection methods — pointers only for now.

Slide 15 — Collapsibility (12 min) · do not drop

- **Start with a two-minute refresher** on the four measures, on one worked example (65% vs 35% died): RD 0.30, RR 1.86, OR 3.45, and what an HR is. Many SPPH students last met the odds ratio in a first-year course; the collapsibility point lands only if the OR is familiar.
- **Definition:** a measure is collapsible if, with no confounding, the whole-population value is a weighted average of the stratum values. **RD and RR are collapsible. OR and HR are not.**
- A hypothetical, in MASLD vocabulary. Age is **balanced by design** — half of each age stratum has a high WHtR — so age is not a confounder. The risks are deliberately large so the effect is visible.

Stratum (n)	Risk, high WHtR	Risk, low WHtR	OR	RR	RD
Age $\geq$ 60 (200)	0.80	0.50	4.00	1.60	0.30
Age < 60 (200)	0.50	0.20	4.00	2.50	0.30
Everyone (400)	0.65	0.35	3.45	1.86	0.30

- Read the OR column. It is 4 in both strata and 3.45 in everyone, and no weighted average of 4 and 4 is 3.45. Adjusting for age moves the OR from 3.45 to 4.00, a **16%** change: change-in-estimate calls age a confounder, and it is not one.
- The RD column does not move. The RR differs by stratum (1.60, 2.50) but the whole population’s 1.86 lies between them, a weighted average. Note in passing that the OR is homogeneous here while the RR is not: whether an effect “varies” depends on the scale. That is Week 5.
- **Two corrections to common slogans.**
 - Non-collapsibility is **not bias**: 4.00 and 3.45 are correct answers to two different questions.
 - It **does not happen under the null**. If the exposure does nothing in any stratum, every OR is 1. It needs a non-null effect and a covariate that predicts the outcome. The previous version of this deck said the HR moves on adjustment “even under a null”; that is withdrawn — the correct phrase is “**even without confounding**”.
- **Size and direction.** It grows with the effect and with how strongly the covariate predicts the outcome — and it moves the adjusted OR or HR **away from 1** (toy: 3.45 $\rightarrow$ 4.00; Lab 3: 1.058 $\rightarrow$ 1.072). Our locked estimate moved **toward 1** (2.34 $\rightarrow$ 1.05), so non-collapsibility explains **none** of that drop; with the Model 1 HR at 1.05 the marginal HR would lie between 1 and 1.05, so if anything it offsets a sliver. The drop is the other mechanisms on Slide 13, and which of those, Slide 8 says the data cannot tell. (An earlier draft said non-collapsibility explained “a small slice” of the drop — wrong direction; withdrawn.)

- The HR also carries a built-in selection over follow-up time: later risk sets hold only the survivors (Hernán 2010).

Slide 16 — Conditional versus marginal, and how to get the marginal one (10 min) · do not drop

- **Conditional effect:** within levels of the covariates — what a regression coefficient is. **Marginal effect:** the population average — what a trial estimates, and usually what a policy question needs.
- **Standardisation (g-computation), four steps:**
 1. Fit an outcome model with exposure and covariates.
 2. Copy the data, set everyone to exposed, and predict each person’s risk.
 3. Set everyone to unexposed, and predict again.
 4. Average each set of predictions and contrast them.

This is L3 Problem 2(c): `predict(model, newdata = everyone_treated, type = "response")`. L3 does it once per measure, with a different model each time: a logistic model for the OR, a Poisson model for the RR, a Gaussian model for the RD. Standardising the logistic model alone would also be valid and gives slightly different RR and RD values; neither is wrong.

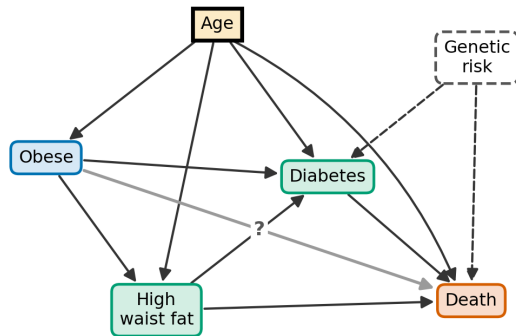
- **The decomposition to carry into the lab:**
 - **Crude vs marginal** is confounding — given that the model is right and the set sufficient.
 - **Marginal vs conditional** is non-collapsibility. It is **zero by construction** for an RR from a log-link model, or an RD from an identity-link model, without product terms: every person’s predicted ratio (or difference) is the same, so the average is too. That equality is algebra, not evidence.
 - On the Slide 15 toy: crude 3.45 = marginal 3.45 (no confounding); conditional 4.00.
- **L3’s three models:**
 - **Logistic** → OR.
 - **Poisson with a log link and robust (sandwich) SEs** → RR, the “modified Poisson” of Zou (2004). The Poisson variance is wrong for a 0/1 outcome; the robust SE repairs the inference.
 - **Gaussian with an identity link and robust SEs** → RD, the linear probability model.
 - Both can predict “risks” outside 0–1 (in Lab 3: Poisson up to 1.57, the linear model from –0.25 to 1.16). That is a known price of these models, not a coding error.
 - Naimi & Whitcomb (2020) Table 2 is the one-page reference.
- **Back to MASLD.** Our 2.34 → 1.05 is crude versus *conditional*. Non-collapsibility would push a conditional HR away from 1, so it cannot account for this drop (Slide 15). The survival analogue of standardisation is a standardised survival curve, or six-year risk under “everyone IV” against “everyone III”; Week 5’s model-based absolute risks start that.

Part 5 — Reporting what you estimated · 13 minutes across the slides below

EpiMethods Concepts: “The Table 2 Fallacy”; “Interaction vs. Effect Modification”.

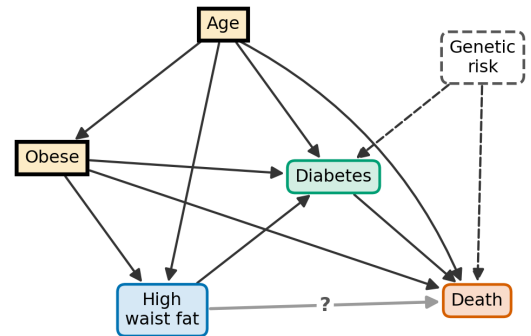
Slide 17 — The Table 2 fallacy: four questions, four DAGs (8 min)

Q1 · Effect of obesity



Adjust: age. Not waist or diabetes — both are on obesity's path

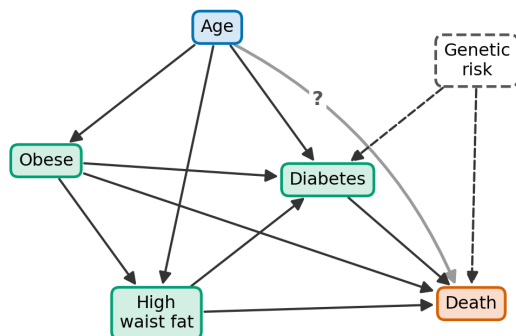
Q2 · Effect of high waist fat



Adjust: age AND obesity (a confounder here). Not diabetes

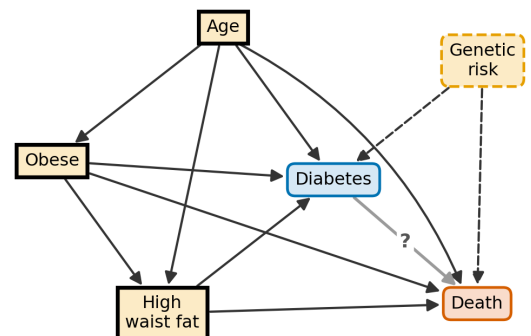
Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

Q3 · Effect of age



Adjust: nothing — nothing in this DAG causes age

Q4 · Effect of diabetes



Adjust: age, obesity, waist — AND genetic risk (unmeasured)

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

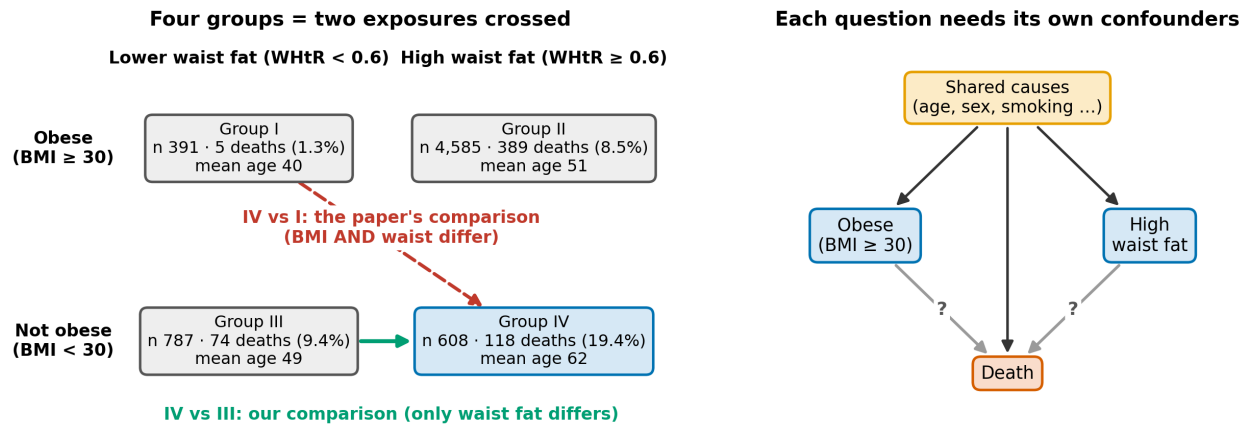
- **Show one model first**, as a paper's Table 2 would print it: obesity + high waist fat
 - age + sex + diabetes, on 5,911 MASLD adults (a separate teaching example that uses *both* body-shape measures). Its rows: obese 0.80 (0.65–0.99), high waist fat 1.17 (0.87–1.57), age per 10 years 2.30 (2.12–2.49), diabetes 1.65 (1.39–1.97). Take a vote: “so obesity is protective?”
- **Then four questions on one DAG**, each with its own adjustment set (checked with dagitty):
 - **Q1, obesity** — adjust for age; not waist or diabetes, which lie on obesity's path.
 - **Q2, high waist fat** — adjust for age *and obesity*, which is a confounder here. This is the paper's own “independent of total body weight” question.
 - **Q3, age** — adjust for nothing.
 - **Q4, diabetes** — needs genetic risk, which is unmeasured.
- **The same variable changes role with the question.** Waist fat is a mediator in Q1 and the exposure in Q2; obesity is the exposure in Q1 and a confounder in Q2.
- **Each question's own model** (with sex, race/ethnicity, smoking and activity added):
 - obesity **0.95** (0.78–1.14);
 - waist **1.24** (0.93–1.67);
 - age **2.38** (2.21–2.56);

- diabetes **1.71** (1.43–2.04), still confounded.

The one-model obesity row (0.80) differs most, because it holds waist and diabetes fixed. The other rows are close — the fallacy is about **meaning**, not only size.

- **Your turn:** flip the obese → waist arrow. Q1 and Q2 swap roles — the DAG, not the data, decides.
- **Credit the paper.** Its Table 2 reports only its exposure rows, with the covariates in a footnote — the recommended layout. Rule: **one exposure, one model, one row reported** (Westreich & Greenland 2013).

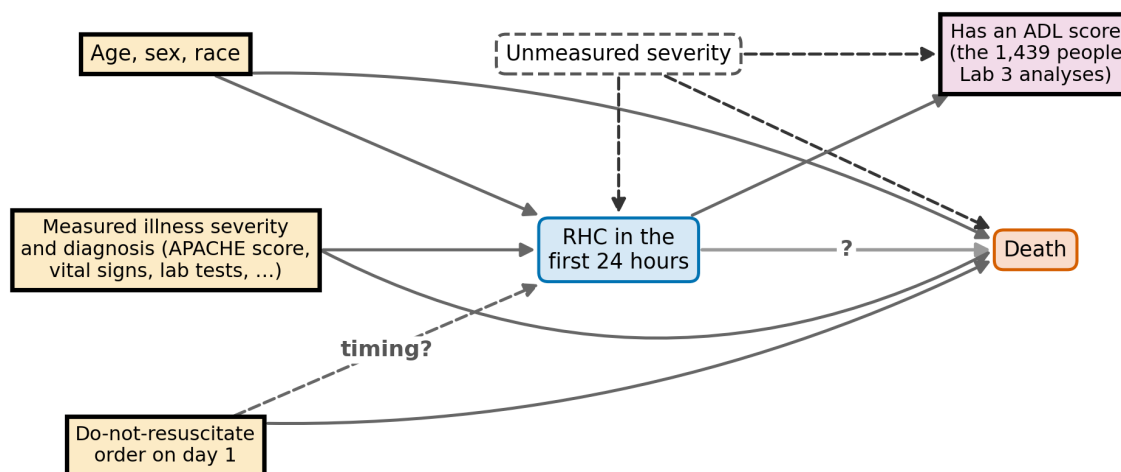
Slide 18 — The four groups are two exposures (5 min)



- Groups I–IV are BMI ≥ 30 crossed with WHtR ≥ 0.6. That makes the phenotype a **joint exposure** — exactly the “joint variable model” of EpiMethods `confounding9`.
- One fitted model, several questions:
 - **IV vs III** — WHtR within the non-obese. This is our locked contrast: a *simple effect*.
 - **II vs I** — WHtR within the obese.
 - **IV vs I** — the paper’s contrast. Both axes change, and the reference cell is obese: Group I is not the doubly-unexposed cell.
- **Week 4’s consequence: the adjustment set follows the question.**
 - For the effect of WHtR within a BMI stratum, you need the confounders of WHtR → death.
 - For the joint effect of both, you need the confounders of **both** exposures (EpiMethods `confounding0`, *Implications for confounding control*).
- Whether WHtR’s effect differs between the obese and the non-obese (II-vs-I against IV-vs-III) is an **effect-modification question**, for Week 5 and P3. Flag the constraint now: every contrast that touches Group I rests on **5 deaths** in the full file, and fewer still on the rows the hazard ratios use.

Part 6 — Thursday · 15 minutes across the slides below

Slide 19 — L3 is RHC, not MASLD (7 min) · do not drop



- **Open by naming it** (the second time). The lab is a different dataset with a binary outcome and a different exposure: right-heart catheterisation in the first 24 hours (`swang1`), and death (`death`).
- **The analytic file** is complete cases: **1,439 of 5,735**. The drop is **entirely one variable**: the ADL score (`adld3p`) is missing for exactly 4,296 patients.
 - Having an ADL score mostly means **surviving the first days**. Thirty-day mortality is **8%** among patients with one and **42%** among those without.
 - The crude OR is **1.25** on all 5,735 rows and **1.04** on the 1,439, before any adjustment. The complete-case step is Slide 6’s lesson — selection is conditioning — in L3’s own data.
 - L3’s crude-versus-adjusted comparison is still on one row set, so it is internally clean; it describes patients who survived long enough to be assessed. Name it and park it for Week 7.
 - To check it yourself: `tapply(rhc$dth30 == "Yes", is.na(rhc$adld3p), mean)`.
- **The adjustment set** is `swang1 + .`: the 22 covariates the exercise kept. `rhc.csv` holds more baseline variables (insurance, income, education, weight, and others) that L3 leaves out. That is the pre-treatment idea from Slide 10, on a pre-chosen list.
 - Ask the room which covariates sit at the **timing edge**, recorded in the same first day as the decision to catheterise: `dnr1` (a DNR order on day 1), `surv2md1` (a model-predicted survival probability), and the day-1 physiology.
 - `dnr1` returns next week as L4’s effect modifier.
- **The tools, and three traps**:
 - `tableone::CreateTableOne` for Table 1; `glm()` with `binomial`, `poisson` and `gaussian` families.
 - `Publish::publish(..., confint.method = "robust")` for the robust SEs. **Install Publish before Thursday** (`install.packages("Publish")`, capital P). The setup page listed it for Lab 4, but L3 needs it too.
 - Publish labels the Poisson column **“HazardRatio”**. It is a risk ratio; the label is the package’s, not an HR.
 - `predict(newdata = ...)` for standardisation. The marginal estimates carry no CIs; a bootstrap would supply them.

Slide 20 — Read the 3×3 , and read 2(e) critically (6 min)

- The exercise page already prints the filled 3×3 in Problem 2(d), so there is nothing to keep secret. **Make the task explaining the pattern, not predicting the numbers:**

	OR	RR	RD
Crude	1.038	1.019	0.009
Conditional	1.072	1.030	0.013
Marginal	1.058	1.03	0.013

- Crude $\rightarrow$ marginal is the confounding the 22 covariates remove: $1.038 \rightarrow 1.058$.
- Marginal $\rightarrow$ conditional in the OR column ($1.058 \rightarrow 1.072$) is non-collapsibility. It is small because the effect is near the null — Slide 15’s size rule, observed.
- The RR and RD rows agree because Slide 16 says they must.
- **Read Problem 2(e)’s printed answer critically.** Three of its sentences need correcting, and the TA should raise them at 2(e):
 1. The OR does **not** move “because the odds ratio is sensitive to how covariates interact with the exposure”. The model has no interaction; Slide 15’s toy moves with none.
 2. Equal conditional and marginal RRs are **not** evidence that the RR is collapsible — the log link with no product terms forces the equality.
 3. The marginal estimates **do** use the covariates; they are standardised over them. It is the crude estimate that ignores them.
- **Logistics:**
 - In the downloaded Rmd, **uncomment the read.csv line** first; the data are in the zip’s Data/ folder.
 - The Rmd’s “Lab Assignment #1” refers to another course: the wrangling code is given, and you have not seen RHC before.
 - Ignore its percentage weights. The course grade is Complete / Incomplete: knit to PDF or HTML, put the group name and present members’ first names in **author:**, add **sessionInfo()**, and submit before the session ends.

Slide 21 — P2: the same judgement, on your paper (2 min)

- **Table 1 by exposure** (including the paper’s p column), a **variable-role table**, and **crude versus adjusted on the same rows**.
- The adjustment set comes from a DAG and the Slide 10 criteria — not from change-in-estimate and not from stepwise selection. For an HR or OR, say that non-collapsibility is one reason the estimate moves.
- P2 is optional and ungraded, with nothing to submit; it enters M1. Say so plainly.
- The M3 question to prepare for: *“Age differs hugely across your groups. Is it a confounder or is it on the causal path — and what would controlling for it do if you’re wrong?”*

Close · 4 minutes

Slide 22 — What adjustment cannot fix; key takeaways; where to read more (4 min)

- **Unmeasured confounding**, and a **wrong DAG**.
- **Selection into the cohort ($FLI \geq 60$)**. In principle it is blockable by adjusting FLI’s inputs, but TG and GGT are plausibly effects of central fat, so doing so trades one bias for another (Week 5 Slide 18, M1).
- **The survey design**. Week 6 runs the analysis design-aware (WTSF2YR/6, strata, PSU). Weighting moves IV vs III from 1.05 to **1.29 (0.89–1.86)** with Model 1 on the 5,939 rows (P4 exemplar), and from 1.09 to **1.43 (0.98–2.07)** with set L on the 5,911 rows (Week6_design_aware.R). Both intervals still include 1.
- **Heterogeneity across subgroups** (Week 5).
- Exit question: *write down the one arrow in your own paper’s DAG you are least sure of.*
- **Key takeaways** (one slide, six lines): assumptions as a DAG; adjust confounders, not mediators, colliders or instruments, and remember selection counts; the modified disjunctive rule when the DAG

is incomplete; ORs and HRs move without confounding; one exposure per model; $2.34 \rightarrow 1.05$ cannot be pinned on one cause.

- **Where to read more** (one slide): the reading map below, with the EpiMethods videos and the EpiMethods quiz for self-testing.

Slide 23 — References (uncounted)

If you are running behind

Drop or shrink in this order. The first two cost the least.

1. **Slide 7 — paradoxes that are colliders (5 min).** Keep the locked-contrast bullet (Group III's higher TG / GGT) as one sentence on Slide 6.
2. **Slide 18 — the joint exposure (5 $\rightarrow$ 2 min).** Show the figure; say “the adjustment set follows the question” and “effect modification is Week 5”.
3. **Slide 3 — explain or predict (4 min).** Fold it into one sentence at the top of Slide 4.
4. **Slide 17 — the Table 2 fallacy (8 $\rightarrow$ 5 min):** show the one-model table and the four DAGs; skip the comparison table. **Slide 14 — selection methods (4 $\rightarrow$ 2 min).** Shrink, do not drop: both are core EpiMethods content. Keep the rule *one exposure, one model, one row*, and *confounding is not a significance test* plus the post-double-selection pointer.

Never drop Slides 8, 15, 16 or 19. The measurement blend, collapsibility, the conditional / marginal decomposition, and the lab's real dataset are what Thursday depends on; everything else supports one of the four.

Reading map — each segment and where students can review it

Segment	EpiMethods page	Key reference
Potential outcomes, DAG grammar (Slides 4–6)	confounding0 (video lessons), DAGcodes	Greenland, Pearl & Robins 1999; Tennant et al. 2021
Confounder, mediator, collider by simulation	confounding1, confounding2, confounding3	—
Simpson's and birth-weight paradoxes (Slide 7)	confounding3b	Hernández-Díaz, Schisterman & Hernán 2006
Instruments and Z-bias (Slide 9)	confounding4	—
Empirical criteria (Slide 10)	confounding0	VanderWeele & Shpitser 2011; VanderWeele 2019
Change-in-estimate (Slide 13)	confounding6	—
Stepwise, purposeful selection, ML (Slide 14)	confounding0	Heinze, Wallisch & Dunkler 2018
Collapsibility, marginal effects (Slides 15–16)	confounding5	Naimi & Whitcomb 2020
Table 2 fallacy (Slide 17)	confounding0	Westreich & Greenland 2013
Joint exposure (Slide 18)	confounding9	Knol & VanderWeele 2012
L3 (Slides 19–20)	confoundingE, confoundingEsolution	Zou 2004

Appendix — three DAGs to paste into dagitty.net

Paste each block into the *Model code* pane at dagitty.net. The “Causal effect identification” panel then gives the adjustment sets each slide states.

Slide 8 — the locked contrast. L stands for age, sex, race/ethnicity, income, smoking and physical activity; Blend is the composite comorbidity indicator. With **Before** and **After** latent, dagitty reports that no adjustment set exists; remove **Before**'s latent tag and it returns {Before, L}.

```
dag {
  WHtR [exposure]
  Death [outcome]
  Before [latent]
  After [latent]
  L -> WHtR
  L -> Death
  L -> Before
  Before -> WHtR
  Before -> Death
  Before -> After
  WHtR -> After
  After -> Death
  WHtR -> Death
  Before -> Blend
  After -> Blend
}
```

Slide 17 — the Table 2 fallacy, four questions on one DAG. Set each variable in turn as the exposure (click it, then “exposure”): - **Obese** — dagitty returns {Age}; - **Waist** — {Age, Obese}; - **Age** — the empty set; - **Dm** (diabetes) — no set, because U (genetic risk) is latent.

```
dag {
  Obese [exposure]
  Death [outcome]
  U [latent]
  Age -> Obese
  Age -> Waist
  Age -> Dm
  Age -> Death
  Obese -> Waist
  Obese -> Dm
  Obese -> Death
  Waist -> Dm
  Waist -> Death
  Dm -> Death
  U -> Dm
  U -> Death
}
```

Slide 19 — L3 with selection on having an ADL score. L stands for the measured covariates. Click ADLrec to condition on it and watch $RHC \rightarrow ADLrec \leftarrow U \rightarrow Death$ open.

```
dag {
  RHC [exposure]
  Death [outcome]
  U [latent]
  L -> RHC
  L -> Death
  U -> RHC
  U -> Death
  U -> ADLrec
}
```

```

RHC -> ADLrec
RHC -> Death
}

```

References

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